Table I. – Chief Features of Nozzles Used in Operational High-Thrust Liquid Rocket Engines (concluded)

Engine	D_t, in.	A_t, $in.^2$	Length, throat to exit, in.	Exit type	Divergence half-angle, deg	D_e, in.	A_e, $in.^2$	A_e/A_t	Nozzle extension	Extension/ nozzle joint	Method of cooling nozzle and extension
RS-1801	4.578	16.5	35.27	72% bell	–	30.9	750	45.6	None	NA	Film and ablation
AJ10-118	5.25	21.64	23.1	Complex contour	–	23.5	432.8	20	Optional	Clamped flange	Regenerative to 20:1; radiation to 30:1 or 40:1
AJ10-138	7.48	43.9	49.61	Bell	–	47.33	1758.5	40	Yes	Bolted flange	Ablation to extension; radiation thereafter
VTR-10	8.320	54.34	62.11	72.3% bell, round entrance	–	57.26	2574.0	47.5	Yes	–	Ablation to 16:1; radiation to 47.5:1
RL10A-3-3	5.14	20.75	46.3	Bell	32.5	38.8	1182.75	57	None	NA	Regenerative
YLR81-BA-11	4.670	17.13	39.8	Bell	25.7	31.326	770.32	45	Yes	Bolted flange	Regenerative to 13:1; radiation to 45:1
AJ10-137	12.447	121.675	111.8	Bell	–	98.40	7605	62.5	Yes	Bolted flange	Ablation to 6:1; radiation to 62.5:1
YLR99-RM-1	8.64	58.6	–	Conical	20	27.20	574.3	9.8	None	NA	Regenerative
YLR113-AJ-1	15.25	182.56	35.41	Conical	17	36.41	1040.66	5.7	None	NA	Film
LR91-AJ-5	9.12	65.41	72	Bell	17	64.0	3217	49.2	Yes	Bolted flange	Regenerative to 13:1; ablation to 49.2:1
LR89-NA-7	16.2	206.0	58	100% bell	–	45.83	1648.8	8	None	NA	Regenerative
LR79-NA-11	16.2	206.0	58	100% bell	–	45.8	1646.6	8	None	NA	Regenerative
H-1C and H-1D	16.16	205	55	Bell	–	45.49	1630	8	Yes	Bolted flange	Regenerative
LR87-AJ-5	15.25	182.56	42.46	Bell	23	43.14	1461.6	8	None	NA	Film and regenerative
J-2	14.7	169.6	90	Bell	4.226 min 27.5 max	77	4656.6	27.5	None	NA	Combined fluid and regenerative
F-1	35	961.4	158	Bell	–	140	15 400	16	Yes	Bolted flange	Regenerative to 10:1; turbine exhaust to 16:1

NA = not applicable